AIRSTREET · FEASIBILITY ASSESSMENT

AirStreet feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-30	Amish Chadha	PR-004 wording follows decision 4 of 2026-10-30
0.1	2026-10-24	Amish Chadha	First feasibility assessment (kit 1.9.1 upgrade)

AirStreet feasibility assessment

AirStreet is feasible to build and run as a prototype: every part is printed, cut from standard aluminium or bought, the power and radio budgets close on FieldNode's shared core, and the science holds (AST-STM-001); what is not yet shown is how accurate the sensors are on a real street after calibration, which only a collocation with a reference station at TRL 4 can show. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	The working principle (electrochemical NO2, laser-scattering PM, passive shield, field calibration by collocation) is established in the literature and in community networks; the constructable model passes 77 constructability checks and has no overlapping parts	High	PM2.5 above 85 % RH (PR-001), field NO2 accuracy (PR-002) and EU limit classification near 40 ug/m3 (PR-004) are at risk	AST-STM-001, AST-CAL-001, `cad/fidelity.json`
Materials	Printed ASA (UV stable), 6063 aluminium bar, angle and sheet, stainless bands and fasteners, brass inserts	High	Creep of the printed saddles under band preload in summer heat is plausible, to check at TRL 4 (S25)	AST-STM-001 S25, BOM
Manufacturing and assembly	A 3D printer with a 250 mm class bed, a saw, a drill and hand tools; 16 assembly steps in the build plan	High	Heat-set inserts need a soldering iron and a steady hand; the pod is the service exchange unit, so spare pods double the printing	AST-BLD-001
Power and controls	87.0 mW design sensor load inside FieldNode's 100 mW allowance; 6.6 days without sun; the NO2 bias held continuously on port B	High	Charging stops below 0 degC (S4); cold spells longer than 4.6 days at -20 degC would empty the cell	AST-CAL-001 A
Data and radio	20-byte records every 5 min on a private gateway or 15 min on The Things Network; within EU868 and US915 rules	High	The radio band follows the partner (AST-DEC-001); range at SF9 in US915 to check in the field	AST-CAL-001 B, AST-REG-001 R1 to R3
Calibration route	CalRig for T, RH and PM; 14-day collocation and every 6 months for NO2	Medium	Needs a partner with a street-level reference station (AST-DEC-001); the street transfer bias is assumed (2 ppb)	AST-CAL-001 C, AST-PRC-001
Supply of parts	SPS30 and SHT45 priced at retail (2026-09-25); NO2-B43F class cell and its front end estimated, not quoted	Medium	The NO2 cell and front end are single-source in practice; prices are indicative until quoted by a distributor	BOM, AST-MKT-001
Regulatory	Indicative data only; radio rules met on paper; pole owner's permission needed per site	Medium	The 2030 EU NO2 limit of 20 ug/m3 changes what PR-004 classifies against (decided on 2026-10-30: both 40 and 20 ug/m3)	AST-REG-001

Ratings: High, Medium, Low.